

Natural Convection Airflow in an Open-ended Isothermally Heated Building Shaft

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ABSTRACT

In this study, the objective is to assess the feasibility of ventilating urban street canyons using building shafts having the external surface of their walls maintained at a temperature higher than the ambient temperature. The aim is to induce airflow by natural convection in an open-ended, isothermally heated building shaft whose length, width, and height are $3\text{m} \times 3\text{m} \times 32\text{m}$ ($9.84\text{ ft} \times 9.84\text{ ft} \times 104.98\text{ ft}$), respectively. The building in which the shaft is installed is an open-ground story building, and the shaft is an open space at the center of the building. A parametric study in which the temperature difference between the walls and the ambient is varied is performed. The six cases of temperature difference covered are 5°C (41°F), 10°C (50°F), 15°C (59°F), 20°C (68°F), 25°C (77°F), and 30°C (86°F). The key variables of interest investigated in this parametric study, are the wall heat flux and the mass flow rate in the shaft. The ultimate aim is to improve the ventilation conditions and increase the mass flow rate in the shaft by modifying the design parameters. This study is a numerical one, where ANSYS-Fluent is used to run the numerical simulations.

INTRODUCTION AND LITERATURE REVIEW

The detrimental impacts of urbanization on human health and the environment have been investigated by many studies, and these impacts include urban overheating and degraded air quality. Aggravated outdoor thermal conditions and increased levels of pollutant concentration have been associated with increased rates of mortality and morbidity, increased energy consumption of buildings, decreased efficiency of power plants, etc. (Santamouris 2020) In 2018, the United Nations estimated that around 55% of the world population resided in urban areas, and this number is projected to increase to 68% by 2050 (World Urbanization Prospects 2018). Consequently, it is essential to abate urban overheating and disperse pollutants at pedestrian level. In this study, the aim is to employ natural convection as a means to ventilate urban street canyons passively in an attempt to upgrade air quality in addition to outdoor thermal conditions.

The different factors that affect the characteristics of natural convection flows have been extensively studied in literature. For instance, (Rahimi, et al. 2018) addressed the impact of changing the enclosure's geometry and the type of fluid used, and (Pandey, GapPark and YeongHa 2019) tackled the impact of adding a body to the enclosure. In another study, (Miroshnichenko and Sheremet 2018) presented the different methods -finite volume, finite difference, Lattice Boltzmann, experiments- that have been employed by researchers to model natural convection flows.

Nusselt number is among the key parameters that characterize the flow and give a better insight on the mode of heat transfer. Knowing that Nu is affected by several factors such as Rayleigh number, Reynolds number, fluid properties, boundary conditions and geometric parameters, many studies have established empirical correlations that

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predict Nu for different flow conditions in ducts. For instance, the model presented by (Sarmientoa, et al. 2020) predicts the Nusselt number in noncircular ducts in microchannel applications. For steady laminar natural convection in an isothermally heated vertical duct, (Yovanovich, Teertstra and Muzychka 2012) developed a correlation of Nu for a wide range of duct shapes and aspect ratios. For forced laminar convection flows, (Muzychka and Yovanovich 2004) established a model for Nu that is valid for a Pr that is greater than 0.1. Other studies reported the variation of the Nu profile. In the studies where isothermal conditions were employed, such as (Ramakrishna, Rubin and Khosla 1981), (Moutsoglou and Park 1993) and (Lakkis and Moukalled 2008), the authors reported that the local value of the Nusselt number decreases with height. Such a profile of Nu is evident when the flow is laminar where a thermal boundary layer near the walls is created. The thermal boundary layer acts as thermal resistance, which reduces heat transfer from the walls to the ambient. Another important parameter in natural convection flows is the Grashoff number, which is the ratio of the buoyancy forces to viscous ones. (Naylor, Floryan and Tarasuk 1991) and (Lakkis and Moukalled 2008) both reported the presence of a recirculation zone at the entrance of the shaft when the Grashoff number is sufficiently high. The investigated parameters in the study conducted by (Sefcik, Webb and Heaton 1991) include the impact of the geometric dimensions of two vents located at the entrance and exit of a duct. Isothermal conditions are applied to one wall, and adiabatic conditions are applied to the remaining three. The results of this study revealed the absence of a recirculation zone when the vents' width is equal to the duct's width. Moreover, the authors reported that Nu increases significantly in the region where recirculation occurs to start decreasing monotonically with height, and this is observed in the cases where the vents' width is smaller than the duct's width. In another study, (Straatman, Tarasuk and Floryan 1993) reported the improvement of the overall heat transfer in an isothermally heated channel as a consequence to adding two adiabatic extensions at the outlet.

Even though thorough work has been done to investigate many aspects of natural convection, the investigation of buoyancy-driven turbulent flow of air in an isothermally heated open-ended building shaft remains limited. The objectives of this study include: (1) investigating the characteristics of three-dimensional natural convection airflow in an isothermally heated open-ended shaft (2) perform a parametric study to assess how the temperature difference, between the walls and the ambient, affects the mass flow rate of air and the overall heat emitted by the walls (3) establish a correlation that predicts the Nu as a function of Rayleigh.

This paper starts with an introduction which describes the motivation, findings of previous studies and the objectives of this study. This is followed by presenting the methods used where a brief description on the geometric parameters, numerical settings and physical settings of the problem is provided. The governing equations in non-dimensional form are then expressed. In the results section, the velocity streamlines at the entrance of the shaft and the variation of the wall heat flux rate with height are analyzed. In addition to displaying the variation of mass flow rate and total wall heat rate as a function of temperature difference, in this section, a correlation between Nusselt and Rayleigh numbers is provided. This paper is closed with a conclusion that recapitulates the main findings.

METHODS

This is a numerical study where ANSYS-Fluent is used given its high reliability in computational fluid dynamics problems. The geometric model consists of a building whose length, width, and height are $25\text{m} \times 25\text{m} \times 36\text{m}$ ($82\text{ ft} \times 82\text{ ft} \times 118\text{ ft}$) respectively. This building has an open ground floor whose height is 4m and comprises an open-ended central shaft whose length, width, and height are $3\text{m} \times 3\text{m} \times 32\text{m}$ ($9.84\text{ ft} \times 9.84\text{ ft} \times 104.98\text{ ft}$) respectively. Since the model is a symmetric one, the study is conducted on one-fourth of the shaft as depicted in Figure 1. The grid used is a hybrid grid, which comprises hexahedral and tetrahedral elements. To resolve the thermal and hydrodynamic boundary layers, inflation layers are added to the shaft's walls. The turbulent model employed is SST k- ω , and ideal gas properties are assigned to air. Figure 1 illustrates the boundary conditions that are assigned to this problem. A temperature equal to 25°C (77°F) is assigned to the far-field planes, which include side planes, top plane and ground. The outer walls of the building, which comprise all surfaces belonging to the building except for the shaft walls, are

adiabatic. Symmetry boundary conditions are assigned to all surfaces that belong to the slicing planes. As for the temperature of the external surface of the shaft wall, it differs from one simulation to the other. In the studied cases, the following temperatures are assigned to the external surfaces of the shaft walls: 30 °C (86 °F), 35 °C (95 °F), 40 °C (104 °F), 45 °C (113 °F), 50 °C (122 °F), and 55 °C (131 °F).

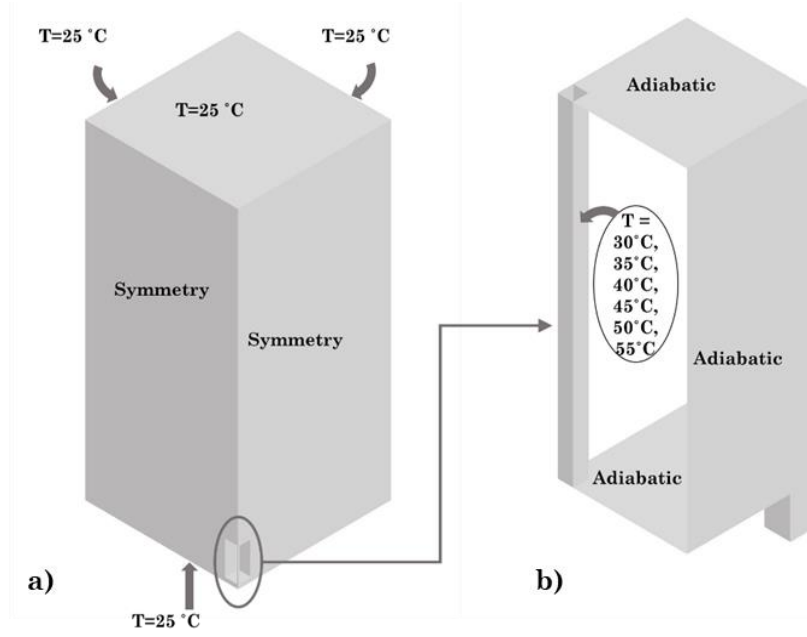


Figure 1: (a) Schematic and boundary conditions of the computational domain (b) Schematic and boundary conditions of the building

GOVERNING EQUATIONS

In this section, the equations of motion that govern natural convection flow are presented, and those equations express conservation of mass, momentum, and energy. In the simulations, the flow is assumed to be steady, three-dimensional, and turbulent with no heat generation. Air is treated as an ideal gas where density is assumed to be variable and the thermal conductivity, dynamic viscosity, and specific heat are assumed to be constants. The constant properties are taken at film temperature, which is the average temperature between the shaft walls and the ambient. Knowing that scaling gives a better insight on the importance of each term in the governing equations, in what follows, the governing equations in non-dimensional form are presented using the following dimensionless variables. In this study, turbulence modeling was performed using the viscous SST k- ω model. The governing equations of the turbulent kinetic energy and specific dissipation rate were developed by (Menter, Kuntz and Langtry 2003).

$$\begin{aligned}\tilde{x} &= \frac{\bar{x}}{b} & \hat{\rho} &= \frac{\rho}{\rho_f} & \tilde{u} &= \frac{\bar{u}}{(gb\beta(T_W - T_\infty))^{0.5}} & \hat{p} &= \frac{p}{\rho_f gb\beta(T_W - T_\infty)} \\ \tilde{\nabla} &= b\nabla & \tilde{g} &= \frac{\bar{g}}{g} & \hat{T} &= \frac{T - T_\infty}{T_W - T_\infty} & Gr &= \frac{\rho_f^2 gb^3(T_W - T_\infty)}{\mu^2} \\ Pr &= \frac{\mu c_p}{k} & \beta &= \frac{1}{T_f} = \frac{1}{0.5(T_W + T_\infty)} & Ra &= Gr \times Pr\end{aligned}$$

- Conservation of Mass

$$\tilde{\nabla} \cdot (\hat{\rho} \tilde{\mathbf{u}}) = 0 \quad 1$$

- Conservation of Momentum:

$$\tilde{\nabla} \cdot (\hat{\rho} \tilde{\mathbf{u}} \tilde{\mathbf{u}}) = -\tilde{\nabla} \hat{p} + \frac{1}{Gr^{0.5}} \tilde{\nabla} \cdot (\tilde{\nabla} \tilde{\mathbf{u}}) + \frac{1}{3Gr^{0.5}} \tilde{\nabla} (\tilde{\nabla} \cdot \tilde{\mathbf{u}}) + \frac{\hat{\rho}}{\beta(T_w - T_\infty)} \tilde{\mathbf{g}} \quad 2$$

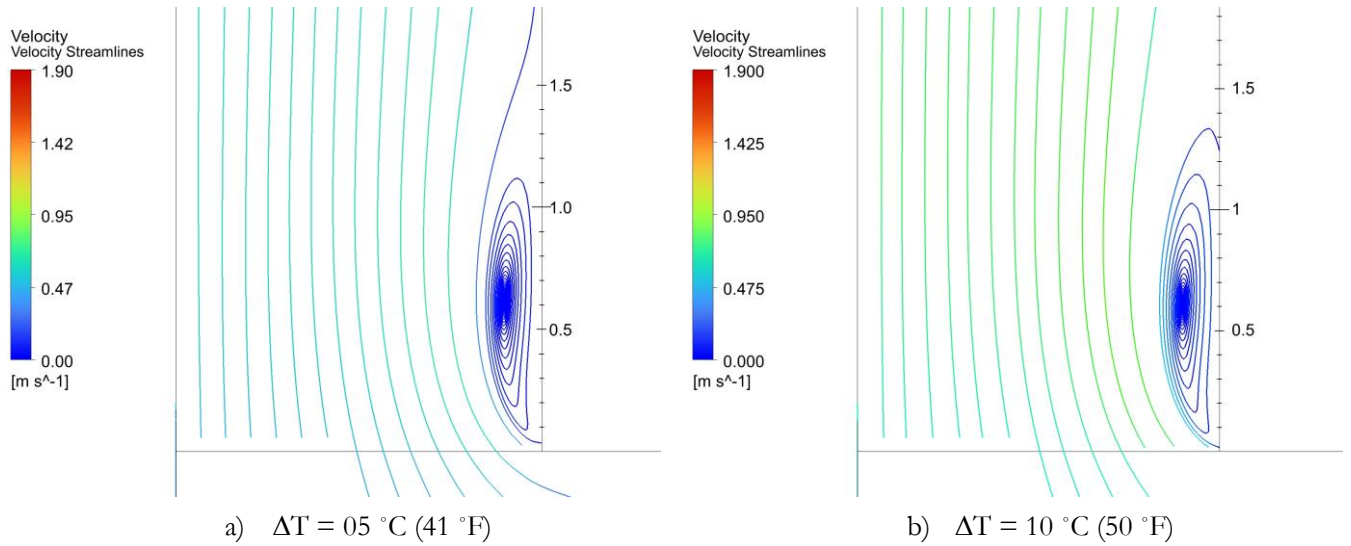
- Conservation of Energy:

$$\tilde{\nabla}(\hat{\rho} \tilde{\mathbf{u}} \tilde{T}) + \frac{T_\infty}{T_w - T_\infty} \tilde{\nabla}(\hat{\rho} \tilde{\mathbf{u}}) = \frac{gb\beta}{c_p} \tilde{\mathbf{u}} \tilde{\nabla} \hat{p} + \frac{1}{Pr Gr^{0.5}} \tilde{\nabla}^2 \tilde{T} + \frac{gb\beta}{c_p Gr^{0.5}} \tilde{\Phi} \quad 3$$

RESULTS

Velocity Streamlines

Figure 2 displays the velocity streamlines as air enters the shaft. Flow separation at the entrance is associated with the decrease in the cross-sectional area through which air is flowing, and it is contingent on Grashoff number as reported by (Naylor, Floryan and Tarasuk 1991) and (Lakkis and Moukalled 2008). If Grashoff number is small, air will enter the channel at low velocities and without any separation. This behavior is not observed in any of the cases investigated in this study. Recirculation at the entrance of the shaft occurs only when Grashoff number is sufficiently high, which is prevalent in the six cases. It is evident that as the temperature difference increases, the intensity of the recirculation zone increases, and air enters the shaft at a higher speed.



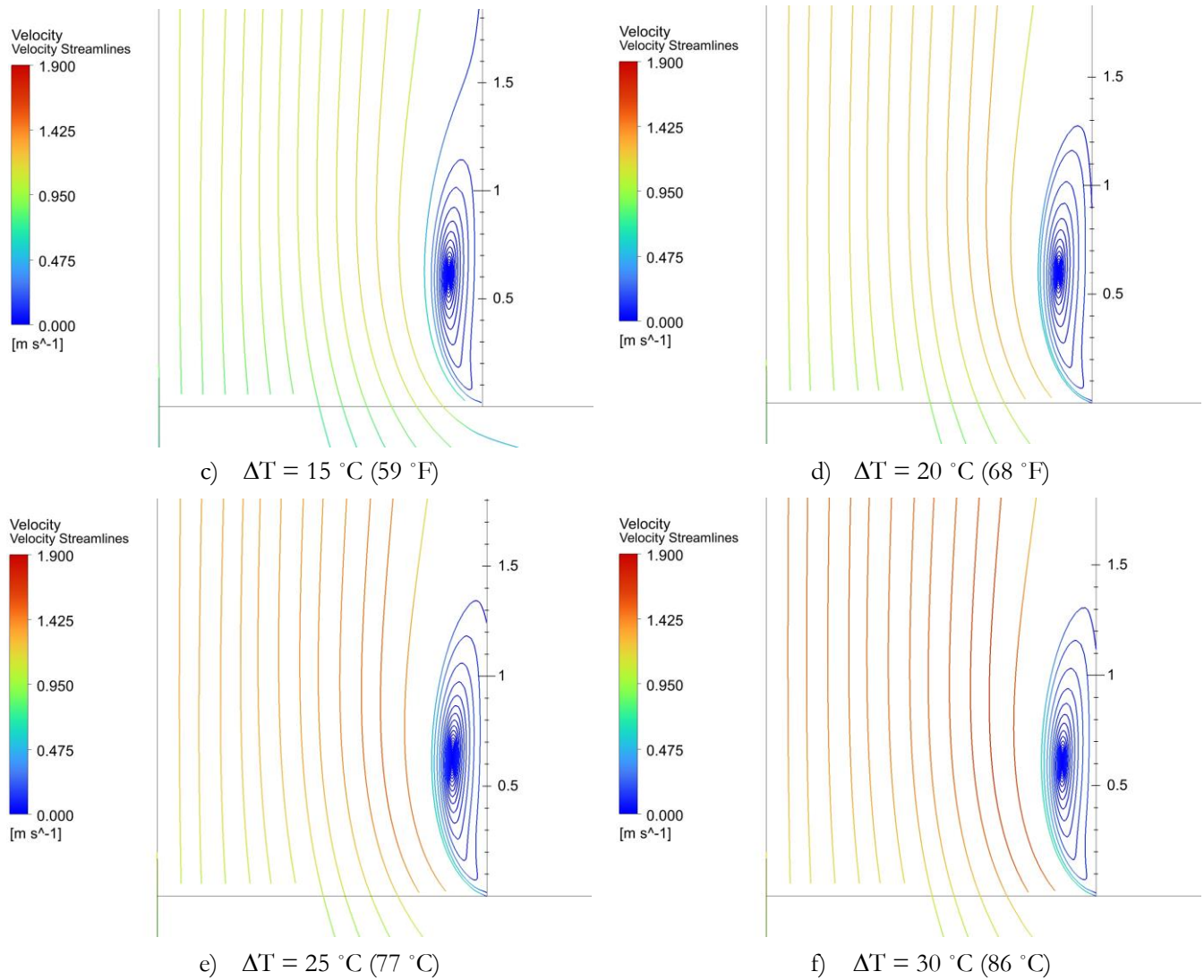


Figure 2: Velocity Streamlines at the entrance of the shaft

Average value of heat flux rate

Figure 3 displays the variation of wall heat flux rate in the shaft for the six temperature difference cases. The temperature and the velocity magnitude of air near the walls are among the factors that affect heat transfer rate where the increase in the air's temperature counteracts the increase in its speed. In other words, as the temperature of air near the walls increases, its tendency to absorb heat decreases, and consequently heat transfer rate is expected to decrease. On the other hand, as the velocity magnitude of air near the walls increases, convection effects are strengthened which results in increasing the heat transfer rate. In this problem, increasing air temperature is coupled with increasing its velocity. Thus, the dominant field - thermal or velocity - could be inferred from the profile of the wall heat flux rate. For all cases, the flow in the shaft could be divided into four regions as shown in Figure 3. In region 1, which corresponds to the region where flow separation occurs, the heat flux rate increases gradually until it reaches a local maximum. The increase in the wall heat flux rate is attributed to the fact that in region 1, air near the walls at higher elevations has a lower temperature than air near the walls at lower elevations. Therefore, the tendency of air to absorb heat as it moves downwards decreases. The onset of the region 2 is at the elevation where the heat flux rate starts decreasing. The decrease of the wall heat flux rate with height in region 2 is attributed to the fact that

air near the walls has not been sufficiently heated, and this renders the velocity field near the walls weaker than the thermal field. Thus, the thermal field dominates the flow in region 2. As for region 3, it is characterized by having an almost uniform wall heat flux rate for the cases of lower temperature difference. For the cases of higher temperature difference, the heat flux in the transition region experiences a slight increase followed by a slight decrease until it reaches a local minimum. The slight variations in the wall heat flux rate in region 3 indicate that the velocity and thermal fields are equally dominant. In region 4, the wall heat flux rate increases steadily with height despite the increase in the temperature of air near the walls, and this implies a dominant velocity field. Figure 3 shows that the profile of the wall heat flux rate becomes steeper for the cases of higher temperature difference. This is attributed to the fact that a higher temperature difference induces a stronger velocity field in the shaft and especially near the walls.

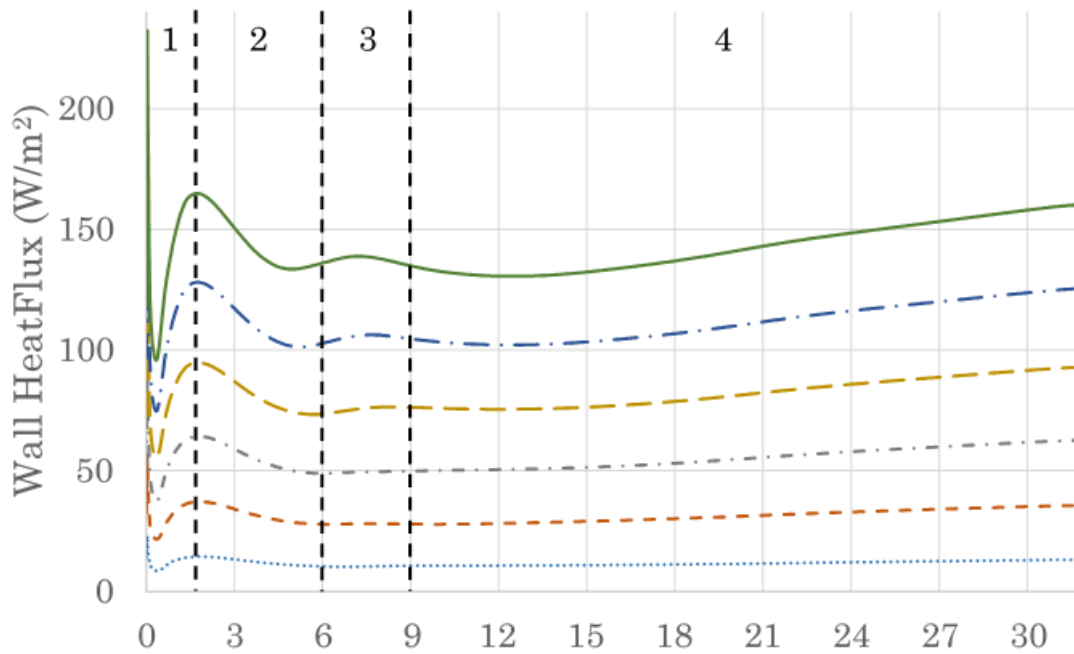


Figure 3: Variation of average wall heat flux with height

Mass flow rate and heat flux vs temperature

Figure 4 displays the variation of the mass flow rate and wall heat flux with respect to the temperature difference between the walls and the ambient. As expected, both parameters increase steadily with temperature difference. It is important to note here that the gradient of the mass flow rate decreases as ΔT increases, whereas the gradient of the total heat rate increases as ΔT increases, and this agrees with the previous findings on the average value of wall heat flux depicted in Figure 3. Figure 4 provides guidelines on inducing airflow in a 3m x 3m (9.84 ft x 9.84 ft) shaft by employing natural convection. For instance, a mass flow rate of 8 kg/s (17.64 lb/s) can be induced in a 3m x 3m (9.84 ft x 9.84 ft) shaft if its walls are maintained at temperature 14 °C (57.2 °F) higher than the ambient temperature. The associated wall heat rate is around 19 kW (64,831 Btu/h)

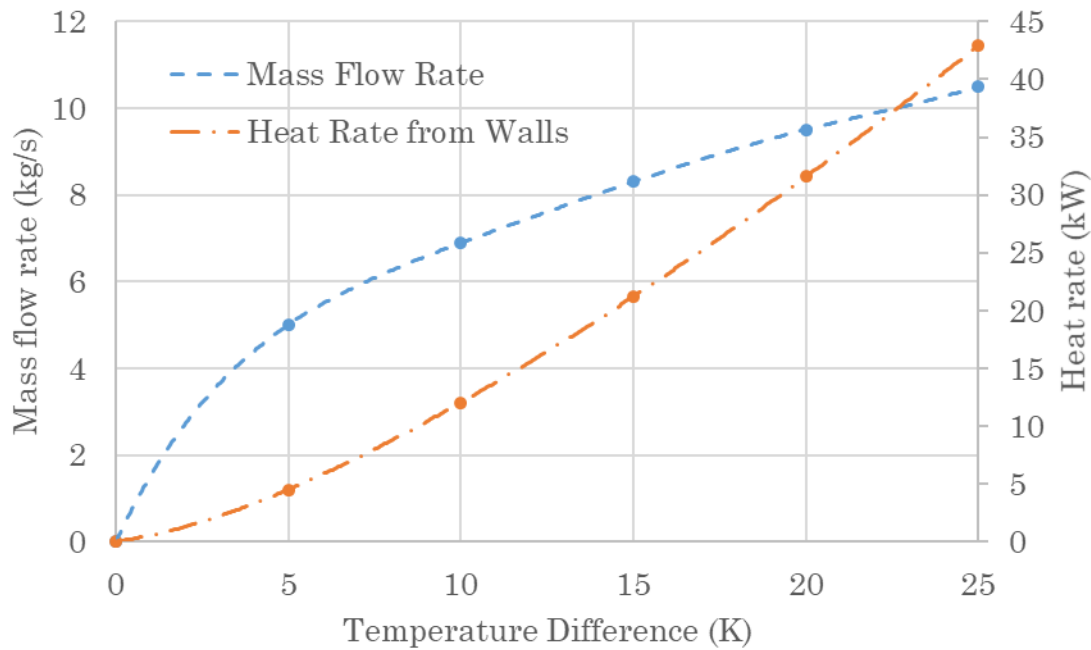


Figure 4: Variation of mass flow rate and wall heat flux with temperature difference

Correlation between Ra and Nu

Nusselt number is an important parameter that allows the determination of the convection heat-transfer coefficient and the total heat flux emitted by the walls. In natural convection problems, Nusselt number is generally correlated with Rayleigh and Prandtl numbers. In Table 1, for each case of temperature difference, the values of mass flow rate, Prandtl, Rayleigh, and Nusselt number are presented. The values of Rayleigh and Prandtl are calculated from the physical settings of the problem, however Nu is based on the results generated in the simulations. Since in this study Pr is an almost constant parameter, in what follows a power law correlation between Ra and Nu is presented.

$$Nu = 0.014784 Ra^{\frac{1}{2.3702}}$$

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Table 1: Mass flow rate, Rayleigh number and Nusselt number for each case of temperature difference

Temperature Difference (°C/ °F)	Mass Flow Rate (kg/s) / (lb/s)	Prandtl Number	Rayleigh Number (x10 ⁹)	Nusselt Number
5 / 41	5.00 / 11.02	0.71	12.604	269
10 / 50	6.89 / 15.19	0.71	24.264	357
15 / 59	8.30 / 18.30	0.71	35.040	417
20 / 68	9.50 / 20.94	0.71	45.000	462
25 / 77	10.50 / 23.14	0.71	54.197	499
30 / 86	11.48 / 25.31	0.71	62.679	530

Conclusion

The objective of this study is to ventilate urban street canyons in an attempt to reduce pollutant concentration at pedestrian level. The different aspects of natural convection flow in an isothermally heated open-ended building shaft are investigated for different shaft wall temperatures. Flow separation is observed at the entrance of the shaft due to the turbulent nature of the flow. According to the profile of the wall heat-flux rate, the flow in the shaft is divided into four zones where each zone is characterized by the interaction between thermal and velocity fields. For the cases covered in this study, the mass flow rate in the shaft ranges between 5-11.48 kg/s (11.02-25.31 lb/s). A correlation that links Rayleigh number to Nusselt number, which ranges between 269-530, is presented.

Nomenclature

$\vec{x}$	Position Vector (m)
$\vec{u}$	Velocity Vector (m/s)
ρ	Density (kg/m ³)
P	Pressure (Pa)
μ	Dynamic Viscosity (kg/m s)
T	Temperature (°C-°K)
b	Width of the shaft (m)
β	Expansion Coefficient (1/K)
$\vec{g}$	Gravitational acceleration (m/s ²)
Φ	Viscous dissipation
Gr	Grashoff number
Pr	Prandtl number
Ra	Rayleigh number
Nu	Nusselt number
c_p	Specific Heat (J/kg K)
ΔT	Temperature difference between shaft walls and the ambient

Subscripts

W	Wall
∞	Ambient
f	Film

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